# Songhyun Bae

*Security Researcher*

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## Profile

I'm a Cyber Defense student at Korea University and a member of CyKor. My experience includes software vulnerability analysis, bug bounty research, and smart contract auditing. Building on this, my current interests include vulnerability discovery, fuzzing, kernel security, Web3 security, AI for security, and AI agent security.

## Education

**Korea University** Mar. 2024 – Present
*B.S. in Cyber Defense*

**Daejeon Daeshin High School** Mar. 2021 – Feb. 2024
*High School Diploma*

## Work Experience

**HashLock** — *Freelance Smart Contract Auditor* Aug. 2025 – Apr. 2026

- Audited Solidity, Vyper, and other Web3 smart contracts for high-severity vulnerabilities.

## Activities

**Web3 Audit Contests** — *Participant* Dec. 2024 – Present

- 40 valid findings across public contests (2 Critical · 18 High · 19 Medium · 1 Low).
- Top placements: 1st Strobe SC (HackenProof), 1st Neutrl Protocol (Sherlock), 3rd Coinbase EIP-7702 Proxy (Cantina), 5th RAAC Core (CodeHawks).
- Full record: Audit portfolio.

**CyKor** — *Member* Sep. 2024 – Present

- Hacking club at Korea University.
- Organizer & challenge author, **CyKor CTF 2025** (Dec. 2025) — authored the Web3 challenge "Multi Chain Vault".
- Organizer & challenge author, **CyKor 2025 Spring Recruit CTF** (Mar. 2025) — authored two Web3 challenges [chal-1] [chal-2].

**KISA Hunting Master, 1st Cohort** — *Participant* Jul. – Aug. 2024

- Performed software fuzzing and vulnerability analysis.
- Discovered vulnerabilities in BandiView, Hauri ViRobot, and SGA ViRUSCHASER.

**Stealien SSL Program, 3rd Cohort** — *Participant* Oct. 2023 – Feb. 2024

- Web-based 0/1-day semi-automated analysis and plugin development.

**KITRI WhiteHat School, 1st Cohort** — *Participant* Sep. 2023 – May 2024

- Automation and enhancement of exploit generation using NLP.

## Talks

**KISA Hunting Master 2nd Cohort** — Special Lecture — Aug. 2025
*Getting Started with Smart-Contract Audit Contests*

**Blockchain Valley** — Security Team Workshop — Mar. – Jun. 2025
*Web3 Security: Smart-Contract Threats & Hands-on Labs*

**27th Hacking Camp** by PoC Security — Aug. 2023
*Hacking with AI: Try Web Hacking with Reinforcement Learning*

## Disclosed Vulnerabilities

**Anthropic Claude Code** — High *(private report)* — HackerOne #3795252

**Apple CoreMedia** — Playback Denial of Service — CVE-2025-24184

**NASA CryptoLib** — Heap-Based Buffer Overflow [Report] — CVE-2025-30216

**Pixar OpenUSD** — Use-After-Free — GHSA-58p5-r2f6-g2cj

**PDFio** — TTF Parser Denial of Service [Report] — CVE-2024-42358

**Toucan Protocol** — Immunefi

- Medium *(private report)* — Immunefi #40684
- Medium *(private report)* — Immunefi #40596
- Low *(private report)* — Immunefi #40497

**Bandiview**

- JXR File Parser — Denial of Service — CVE-2024-45870
- PSD File Parser — Denial of Service — CVE-2024-45871
- PSD File Parser — Heap-Based Buffer Overflow — CVE-2024-45872

**WordPress Plugins**

- Icons Font Loader — Arbitrary File Upload — CVE-2024-24714
- Fast Custom Social Share by CodeBard — Stored XSS — CVE-2023-48329
- Simple File List — Stored XSS — CVE-2023-39924
- wpView — Stored XSS — CVE-2023-33213

**Hauri ViRobot** — DLL Hijacking — KVE-2024-0395

**SGA ViRUSCHASER** — DLL Hijacking — KVE-2024-0410

## Achievements & Awards

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| **5th** | Defense AI Cyber Security Hackathon | 2026 |
| **10th** | LakeCTF Finals 25-26 with CyKor | 2026 |
| **1st** | Cyber Security Hackathon with Boolean | 2024 |
| **Finalist** | CODEGATE Junior | 2023 |
| **Finalist** | WACON Junior with 1등하면전액기부팀 | 2023 |
| **Finalist** | WhiteHat Contest Junior | 2023 |
| **Finalist** | CCE Junior | 2023 |

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| **1st** | Intelligent Mobility SW Contest (AI) with 파송송 | 2023 |
| **1st** | Chungcheong Youth Hacking Defense Contest with 1등 커트라인 | 2022 |
| **1st** | AI Self-Directed Research Competition | 2022 |
| **2nd** | Korea Code Fair Hackathon with Krita | 2022 |
| **3rd** | e-ICON World Contest with Green Energy | 2022 |
| **4th** | KOSPO Web Security Competition with 걸어서 웹 속으로 | 2022 |
| **4th** | K-ICT Data Creator Camp with Finder | 2022 |